



Attribution of value for combination immune checkpoint inhibitors in non-small cell lung cancer[☆]

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ABSTRACT

Immunotherapy represents a significant breakthrough in the treatment of cancer, including non-small cell lung cancer (NSCLC). Immune checkpoint inhibitors (ICIs) are used in combination with other treatments to provide clinically meaningful outcomes for NSCLC patients. However, there are distinct mechanisms of action that an ICI may provide such clinically meaningful benefits. We focused on the valuation of ICIs when used in combination with existing treatments for NSCLC, by addressing the following questions: (1) do combination ICIs improve clinical outcomes due to independent, rather than synergistic or additive drug action; and (2) how should we attribute value to the constituent parts of combination ICIs? To address these questions, we reviewed the United States Food and Drug Administration (FDA) drug database and Clinicaltrials.gov from January 1, 2012, until June 1, 2022, to identify approved indications of combination ICIs in NSCLC. For valuation methods, a separate search was conducted in PubMed, health technology assessment databases, and grey literature to identify published value assessment or attribution methods, specifically in the context of combination (cancer) treatments. As of June 1, 2022, the FDA approved eight combination ICI indications for NSCLC. The underlying mechanisms for the improved clinical benefits of these ICI therapies are not well studied. The superiority of combination ICI therapies compared to monotherapy in multiple indications does not indicate whether synergy or additivity is involved, or necessary.

Policy statement: We encourage further research on the development of value attribution framework methods for combination therapies to quantify their added health benefits and economic value in the future. Given the valuation challenges of combination ICIs, their mechanism of action poses significant uncertainty and requires further clinical investigation to address whether synergy or additivity is existent.

1. Introduction

Immunotherapy represents a significant breakthrough in the treatment of cancer. Immune checkpoint inhibitors (ICI) targeting cytotoxic T lymphocyte-associated protein 4 (CTLA-4) and programmed cell death protein 1/programmed cell death ligand 1 (PD-1/PD-L1) have been integrated into the standard of care for patients with cancer, including non-small cell lung cancer (NSCLC) [1]. To explore the clinical efficacy and safety of these ICIs, thousands of clinical trials are underway [2]. Increasingly, approaches to treat many cancers involve the administration of a number of drugs with distinct yet complementary mechanisms of action in combination or in close sequence, as part of a treatment that

seeks to minimize drug resistance [3]. ICIs that are used in combination with an ICI or other drugs aim to provide clinically meaningful outcomes for cancer patients [4,5]. However, there are distinct mechanisms of action that an ICI may provide such clinically meaningful benefits. For example, when combination ICIs work through *independent* drug action, the benefit to an individual patient is attributed to only one of the drugs in the combination; and the benefit over monotherapy can be due to increasing the odds that the combination includes a drug that is effective for a given patient [6]. This mode of drug action may contrast with *synergy*, where one ICI may enhance the clinical activity of another drug in the combination; and *additivity*, where the clinical benefit is considered to be the sum of multiple drugs in the combination [6].

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Combination ICIs are increasingly used in clinical trials and daily practice, however, the societal challenges that these drugs generate are increasingly becoming evident, especially when their price tags result in payment hurdles in many healthcare systems [7]. Combination ICIs face challenges in the recognition of value by indication, and the complications of attributing value for combination ICIs are also evident in NSCLC. From an economic perspective, many payers or health technology assessment (HTA) bodies do not often find that the expected additional benefits from adding a new ICI to a reimbursed drug or drug combination outweigh treatment costs [8]. Standard approaches to HTA, to determine whether an incremental cost-effectiveness ratio for a drug is deemed acceptable given a willingness to pay threshold (WTP), may sometimes generate results that are counterintuitive, or inconsistent with the societal preferences [9–12]. To address the challenge of the valuation of combination ICIs, it should be possible to adjust the price of both constituents of a combination regimen to reflect the value they offer when used in that combination [13]. Given that the constituent drugs were not developed solely for use in a specific NSCLC indication, this approach would be feasible in practice if indication-specific prices or payment models are permitted.

Therefore, a key problem area of unrealized benefits of ICIs is the difficulty in finding a solution to paying for them when used in combination. Previous research contributed to the debate as to why a new approach is required to address the value attribution problem in combination drugs [8,13,14]. In this study, we focused on the valuation of ICIs when used in combination with existing treatments in NSCLC, by addressing the following questions: (1) do combination ICIs improve clinical outcomes due to independent, rather than synergistic or additive drug action; and (2) how should we attribute value to the constituent parts of combination ICIs?

2. Identification of combination ICIs and valuation methods

We reviewed the United States Food and Drug Administration (FDA) drug database and Clinicaltrials.gov from January 1, 2012, until June 1, 2022, to identify approved indications of combination ICIs in NSCLC. For valuation methods, a separate search was conducted in PubMed, health technology assessment databases, and grey literature to identify published value assessment and/or attribution methods, specifically in the context of combination (cancer) drugs. The searches were built using the Population Intervention Comparison Outcome (PICO) framework (see [Supplementary Appendix](#)). The findings were summarized descriptively. Clinical trial number, combination ICI(s) target or target expression level, histology, approval by the FDA, treatment line or setting, ICI dose, and schedule were presented in a tabular format.

2.1. The clinical landscape of combination ICIs in NSCLC

As of June 1, 2022, the US FDA approved eight combination ICI indications in the treatment of NSCLC. These approved combination ICI indications comprise: (1) “*Pembrolizumab* in combination with pemetrexed and carboplatin for the treatment of patients with previously untreated metastatic non-squamous NSCLC”; [15] (2) “*Pembrolizumab* in combination with pemetrexed and platinum as first-line treatment of patients with metastatic, non-squamous NSCLC, with no epidermal growth factor receptor (EGFR) or anaplastic lymphoma kinase (ALK) genomic tumor aberrations”; [16] (3) “*Pembrolizumab* in combination with carboplatin and either paclitaxel or nab-paclitaxel as first-line treatment of metastatic squamous”; [17] (4) “*Atezolizumab* in combination with bevacizumab, paclitaxel, and carboplatin for the first-line treatment of patients with metastatic non-squamous NSCLC, with no EGFR or ALK genomic tumor aberrations”; [18] (5) “*Atezolizumab* in combination with paclitaxel protein-bound and carboplatin for the first-line treatment of adult patients with metastatic non-squamous NSCLC, with no EGFR or ALK genomic tumor aberrations”; [19] (6) “*Nivolumab* plus ipilimumab as first-line treatment for patients with

metastatic NSCLC whose tumors express PD-L1 ($\geq 1\%$), as determined by an FDA-approved test, with no EGFR or ALK genomic tumor aberrations”; [20] (7) “*Nivolumab* plus ipilimumab and two cycles of platinum-doublet chemotherapy as first-line treatment for patients with metastatic or recurrent NSCLC, EGFR or ALK genomic tumor aberrations”; [21] (8) “*Nivolumab* with platinum-doublet chemotherapy for adult patients with NSCLC in the neoadjuvant setting” [22]. [Table 1](#) shows additional details of all combination ICIs approved by the FDA for the treatment of NSCLC.

2.2. The case of atezolizumab combination therapy in NSCLC

Combinations ICIs may provide clinically meaningful benefits to patients, measured by population-averaged outcomes, without a requirement for pharmacologic additivity or synergy in individual patients. It is helpful to recognize the contribution of each drug as a monotherapy, and also recognize respective contributions to the combination regimen. [Fig. 1](#) depicts a schematic representation of the constituents of combination therapy; including a backbone therapy, an add-on therapy, and (potential) additional drugs. To determine whether independent drug action occurs in ICIs, Palmer et al. [6] utilized a predictive model in a retrospective analysis of 13 clinical trials of combination ICIs representing eight different cancer types. Specifically, in NSCLC, Palmer et al. focused on atezolizumab in combination with bevacizumab, paclitaxel, and carboplatin for the first-line treatment of patients with metastatic non-squamous NSCLC, with no EGFR or ALK genomic tumor aberrations [6].

The IMpower150 clinical trial evaluated first-line treatment of non-squamous NSCLC patients using bevacizumab, carboplatin, and paclitaxel, with or without the PD-L1 inhibitor atezolizumab [23]. In this trial, progression-free survival (PFS) in the ICI combination arm “surpassed the expectation of independence with a hazard ratio of 0.84” ($P = 0.01$, $n = 356$; median PFS surpassed expectation by nine days) [23]. The IMpower150 trial did not evaluate atezolizumab as monotherapy, and the initial assessment used data from the OAK trial on atezolizumab monotherapy in non-squamous NSCLC, with no PD-L1 preselection [24]. However, the OAK trial enrolled patients receiving atezolizumab as second or third-line therapy. The BIRCH trial investigated atezolizumab in NSCLC (72 % non-squamous, all tumors $\geq 5\%$ PD-L1 positive), [25] and reported that atezolizumab was more active as first-line than as second- or third-line therapy.

For the atezolizumab combination therapy, Palmer et al. used a probability model to calculate the expected PFS distribution that would occur if the combination worked through an independent mechanism of action. [6] This expected distribution from independent drug action was then compared to the actual clinical trial results. If the actual PFS observed in the clinical trial was not different from the predicted benefit, then the authors concluded that the combination worked through independent drug action. If the actual PFS was significantly greater than the prediction, then the combination benefit was assumed to be due to synergy or additivity [6]. The authors measured clinically observed differences in atezolizumab activity by the line of therapy, [25] to construct a synthetic arm for atezolizumab in the first-line treatment of non-squamous NSCLC [26]. The authors found that IMpower150 closely matched the expectation of independent drug action (hazard ratio: 1.04, $P = 0.46$, $N = 356$). Palmer et al. indicated that: “atezolizumab is less active at second-line than first-line therapy, and independent drug action may explain the benefit of adding atezolizumab to combination chemotherapy” [6]. The measured differences in atezolizumab activity by line of therapy in the BIRCH trial supported this finding [25]. The authors concluded that “PFS in IMpower150 is most likely consistent with the expectation of independent drug action”. Although these findings are based on a retrospective study design and a model, a more precise assessment of independence in such cases is unlikely, because such an assessment would be unethical, if withholding first-line therapy from eligible patients were needed.

Table 1

Combination immune checkpoint inhibitors approved by the United States Food and Drug Administration for the treatment of non-small cell lung cancer [2017–2022].

Clinical Trial (s) [Number]	Combination Immune Checkpoint Inhibitor [target or target expression level, histology]	FDA Approval	Treatment Line	Dose and Schedule
KEYNOTE 021 [NCT02039674]	Pembrolizumab combination (Pemetrexed + Carboplatin) [PD-L1 TPS $\geq$ 1 %, non-squamous]	May 10, 2017	First-line	200 mg as an intravenous infusion every 3 weeks until disease progression, unacceptable toxicity, or up to 24 months in patients without disease progression
KEYNOTE 189 [NCT02578680]	Pembrolizumab combination (Pemetrexed + Carboplatin/Cisplatin) [regardless of PD-L1 expression, non-squamous]	August 20, 2018	First-line	200 mg as an intravenous infusion over 30 min every 3 weeks
KEYNOTE 407 [NCT02775435]	Pembrolizumab combination (Paclitaxel or Nab-Paclitaxel) [regardless of PD-L1 expression, squamous]	October 30, 2018	First-line	200 mg intravenously every 3 weeks, prior to chemotherapy when given on the same day, until disease progression, unacceptable toxicity, or 24 months after initiation.
IMpower 150 [NCT02366143]	Atezolizumab combination (Bevacizumab, Carboplatin, Paclitaxel) [regardless of PD-L1 expression, non-squamous]	December 6, 2018	First-line	1200 mg intravenously over 60 min every 3 weeks
IMpower 130 [NCT02367781]	Atezolizumab combination (Carboplatin/Nab-Paclitaxel) [regardless of PD-L1 expression, non-squamous]	December 3, 2019	First-line	1200 mg as an intravenous infusion every 3 weeks, prior to chemotherapy when given on the same day
CheckMate 227 [NCT02477826]	Nivolumab + Ipilimumab combination [PD1, CTLA-4, PD-L1 TPS $\geq$ 1 %, squamous or non-squamous]	May 15, 2020	First-line	Nivolumab 3 mg/kg every 2 weeks Ipilimumab 1 mg/kg every 6 weeks until disease progression, unacceptable toxicity, or up to 2 years in patients without disease progression
CheckMate 9LA [NCT03215706]	Nivolumab + Ipilimumab combination (two cycles of platinum doublet) [PD1, CTLA-4, regardless of PD-L1 expression, squamous or non-squamous]	May 26, 2020	First-line	Nivolumab 360 mg every 3 weeks Ipilimumab 1 mg/kg every 6 weeks and 2 cycles of platinum-doublet chemotherapy, continued until disease progression, unacceptable toxicity, or up to 2 years in patients without disease progression.
CheckMate 816 [NCT02998528]	Nivolumab* combination (+ platinum-doublet chemotherapy) [PD1, regardless of PD-L1 expression, squamous or non-squamous]	March 4, 2022	Early-stage, Neoadjuvant setting	Nivolumab 360 mg with platinum-doublet chemotherapy on the same day every 3 weeks for 3 cycles

*Nivolumab platinum-doublet chemotherapy combination represents the first FDA approval of neoadjuvant therapy for early-stage NSCLC.

Abbreviations: NSCLC: Non-Small Cell Lung Cancer, US: United States, FDA: Food and Drug Administration, NCT: National Clinical Trial, PD1: Programmed cell death protein 1, PD-L1: Programmed death ligand 1, CTLA-4: Cytotoxic T lymphocyte-associated antigen 4, TPS: Tumor Proportion Score, TC: PD-L1 stained Tumor Cells, IC: PD-L1 stained tumor-infiltrating Immune Cells.

3. Attribution of value for the constituent parts of combination ICIs

Combination therapies in many disease areas share features that are more difficult to demonstrate economic value under existing valuation frameworks. However, value demonstration has been particularly challenging for combination ICIs, given their complex mechanisms of action. Valuation challenges of combination therapies were previously addressed in a report published by the NICE Decision Support Unit (DSU), which highlighted cases that the combination drugs were not deemed cost-effective, even if the novel add-on therapy is provided at zero cost [10]. The DSU report showed that “even if the price of the add-on therapy is zero, the longer duration of treatment with the backbone therapy increases the total costs of treatment in the combination therapy” [10]. Moreover, it showed that “if a combination therapy increases the amount of time that patients spend in a post-progression state, patients may require additional care, thus incurring more costs” [10,11]. Therefore, clinically beneficial combination therapies may not be cost-effective even when the novel add-on therapy is provided for free [10,11]. Other valuation frameworks that focused specifically on treatments in oncology (i.e., American Society of Clinical Oncology, National Comprehensive Cancer Network, European Society for Medical Oncology, Memorial Sloan Kettering Cancer Center’s Drug Pricing Lab) had not differentiated on their methodology or provided specific methods to address the valuation challenges of combination therapies [27–30].

More recently, a group of international stakeholders and experts in HTA outlined key challenges and potential insights to valuing and paying for combination therapies in oncology [12,13]. The report of these stakeholders and HTA experts indicate that when existing monotherapies are combined, however, their value is not proportional to their combined costs, it would be appropriate for constituent drug prices to be re-assessed and negotiated [12,13]. Similarly, when new add-on treatments are combined with an existing backbone therapy and provide

clinical benefit, it is recommended that the price of the backbone therapy should be re-assessed in its new use [12,13]. However, the latter finding raises implementation issues and legal challenges, especially when the constituent therapies are produced by different manufacturers. Fig. 2 depicts a schematic representation of potential methods to be considered for valuing and paying for combination ICIs.

In 2021, Briggs et al. proposed a conceptual value attribution framework for combination drugs [8]. This conceptual framework provides a structured approach for determining how to attribute the benefit of combination therapy to each of its component therapies [8]. One advantage of the proposed framework is that it is independent of price and focuses solely on the (value of) health benefits (i.e., QALY), a metric used in cost-effectiveness analysis to compare the health benefits of different treatments [8]. Various case examples are employed to demonstrate that proposed solutions could be based on publicly available evidence for treatment combinations, including ICIs. However, the underlying assumption in this framework was that the mechanism of drug action for component treatments would be additive, sub-additive, or synergistic. This is in contrast to the findings of Palmer et al., given that independent drug action may explain the benefit of adding atezolizumab to combination chemotherapy [6].

Briggs et al. discussed potential methods for ICI combination therapies that are characterized by key features of the problem: “perfect/imperfect information about the monotherapy effect of component therapies (individual effects are known/unknown)” and “balanced/unbalanced market power between their manufacturers (unbalanced: one manufacturer already has market access and a new entrant seeks to add to this ‘backbone’ therapy)” [8]. The authors used the term “imperfect” information to define scenarios where the “independent benefit of one or more of the component therapies is unknown for the indication under consideration”. Imperfect information scenarios typically arise when a novel add-on is combined with an existing backbone therapy [8]. In contrast, the authors used the term “perfect” information to define scenarios where the “independent benefit of every component benefit is

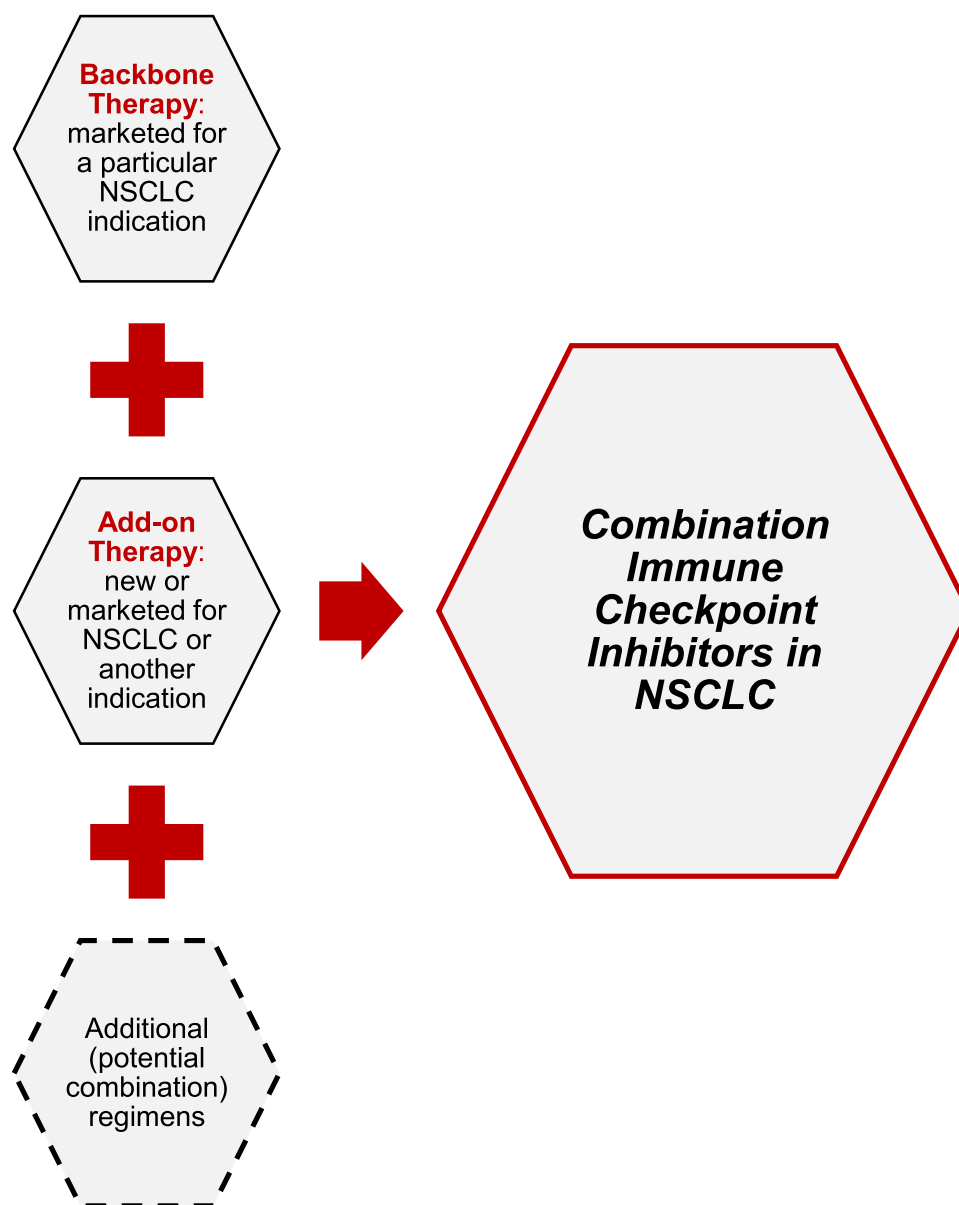


Fig. 1. Constituent parts of combination immune checkpoint inhibitors. Figure Legend: A schematic representation of the constituents of a combination immune checkpoint inhibitor; a backbone therapy, an add-on therapy, and (potential) additional drugs.

known for the indication for which the combination therapy is being assessed" [8]. Perfect information scenarios typically arise when combination therapy is composed of two existing therapies that have already been appraised and approved independently [8]. Briggs et al. introduced the term "imbalanced market power" to define scenarios where "the manufacturer of one component therapy has more control over pricing decisions compared to the manufacturer of another component therapy" [8]. It was indicated that there could be an imbalance of market power when a "novel add-on is combined with an existing backbone therapy that has already been appraised and approved" [8]. Additionally, there can be a market power imbalance when "one of the component therapies holds a larger share of the market, either in the indication for which the combination is being appraised or across multiple indications" [8]. The authors used the term 'balanced market power' to define scenarios where none of the component therapy manufacturers has more control over pricing decisions than others [8]. "Market power can be balanced in cases where a combination therapy consists of two or more existing therapies that have already been appraised and approved", and there is no large discrepancy in their respective market shares [8]. The authors

proposed potential solutions to each of these four scenarios.

One of the advantages of this framework is that it is independent of price and focuses on health consequences (i.e., QALY). Together, this 'net-QALY' represents the "value of a new treatment in health terms and can be converted to a monetary value by multiplying the threshold WTP for a QALY" [8]. This monetized value of the net-health consequences can then be considered to be the maximum (differential) price that the health system would be willing to pay for a combination treatment. Thus, this framework avoids the complications of judging whether the existing price charged for a product is 'fair' and it does not depend on confidential patient access schemes [8].

Similarly, in 2021, researchers at the Office of Health Economics in the UK published a report on the conceptual valuation of combination therapies [14]. The authors focused on the "additive scale" and the "relative change in treatment duration as compared to the increase in overall survival" [14]. This conceptual framework shows that the value attribution approaches, namely "the partial information incremental value approach, and the full information monotherapy ratio approach", are only comparable if an implicit assumption about "additive scale" is

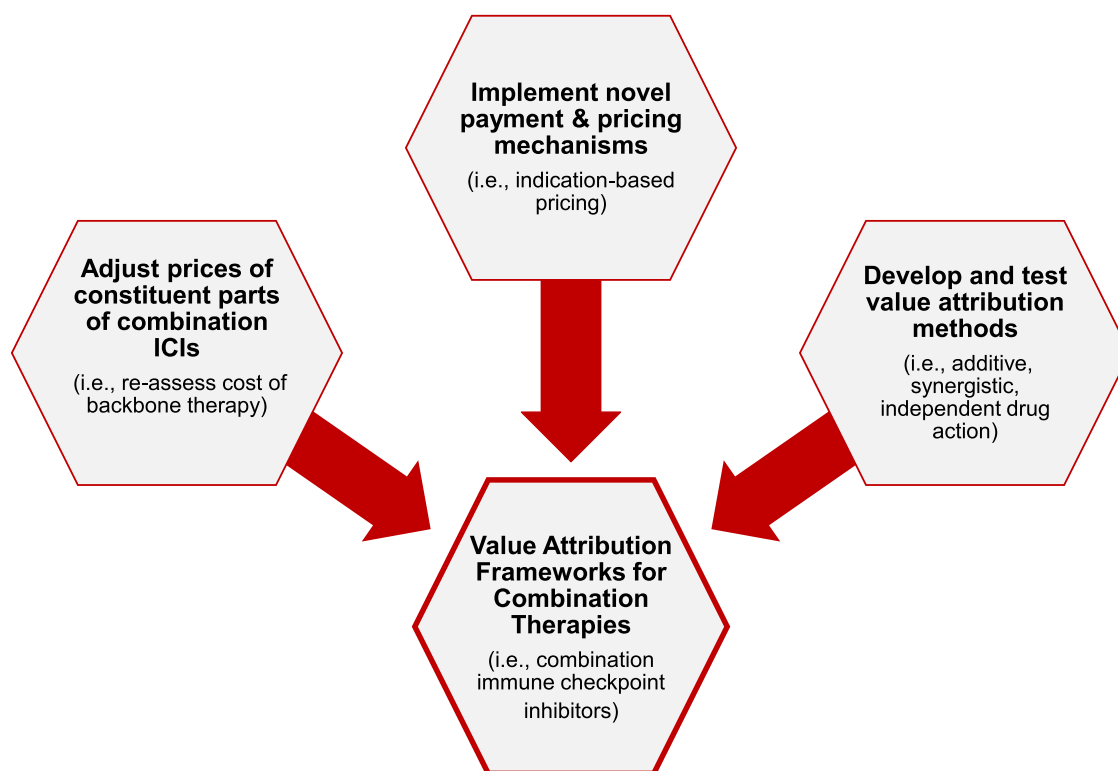


Fig. 2. Valuing and paying for combination ICIs. Figure Legend: A schematic representation of potential methods to be considered for valuing and paying for combination immune checkpoint inhibitors.

made [14]. This assumption is, again, in contrast to the finding that combination therapies may work through an independent mechanism of drug action [6]. Additional considerations should be given to questions related to the evaluation and selection of clinical evidence. These considerations may introduce additional layers of complexity that needs to be taken into consideration. Moreover, the challenges that these potential valuation methods might face in different pricing systems and HTA jurisdictions should be explored.

3.1. The case of ipilimumab therapy in NSCLC

In the first-line treatment of NSCLC, ipilimumab in combination with nivolumab is approved based on the CheckMate 227 [NCT02477826] and the CheckMate 9LA [NCT03215706] trials, for patients with or without (i.e., for PD-L1 $\geq 1\%$) platinum doublet chemotherapy [31,32]. Although ipilimumab's drug action, as a single-agent, was investigated in NSCLC trials (i.e., CA184-041 trial [NCT00527735]), [33], its single-agent activity is not fully understood or demonstrated, because ipilimumab has not been approved by regulatory agencies as a single-agent treatment in NSCLC. Valuation studies focusing on the cost-effectiveness of nivolumab-ipilimumab combination therapy, based on the CheckMate 227 and CheckMate 9LA clinical trials, reported substantial costs (i.e., mostly attributable to drug acquisition costs) [34–40]. In the CheckMate 227-based cost-effectiveness analyses, additional cost outcomes of nivolumab+ ipilimumab combination therapy ranged from US\$ 66,218 [34] to US\$ 201,900 [35]. Similarly, in the CheckMate 9LA-based cost-effectiveness analyses, additional cost outcomes of nivolumab+ipilimumab combination therapy ranged from US\$ 161,993 [37] to US\$ 217,820 [40]. Given these substantial economic consequences, ipilimumab's drug action (as a combination therapy) poses significant uncertainty and requires further (clinical) investigation to address whether synergy or additivity is existent.

3.2. Indication-based pricing for combination ICIs

Indication-based pricing (IBP) can be used to allow differential prices to be set for the same drug when used in distinct indications [41]. In the context of combination ICIs, IBP is relevant not only when drugs have multiple indications, but also where they have different uses within the same indication. ICIs may have different uses within the same indication either as a monotherapy or an add-on as part of a combination. If differential or IBP prices are permitted depending on distinctive uses, there will be greater flexibility in determining the range of acceptable and feasible prices for the component parts of combination ICIs. However, the feasibility of differential drug pricing depends on the local market construct [14]. Although there are various approaches to pricing combination drugs based on their indications: weighted average (blended), differential discounts, different brand names for different indications, outcome-based payment models, among others, in some healthcare markets IBP or the differential prices across indications, is not permitted [42]. For example, if a backbone therapy is approved for multiple indications, and the market does not allow any price or discount variation by indication, the manufacturer(s) may have little incentive to amend prices [8]. “The manufacturer of the add-on therapy could also set a lower price so that the combination therapy will be cost-effective”. In this case, the manufacturer of the add-on therapy captures limited value from the combination therapy [41,43]. “When the backbone therapy is an established therapy and the add-on is a new market entrant, many novel combination therapies may fail to demonstrate cost-effectiveness” [8]. This could have the potential to reduce patient access to clinically important combination ICIs [43].

Alternatively, the lack of price flexibility means that the prices agreed upon for the combination ICIs automatically apply to all other indications of the constituents when sold as monotherapy or within other combinations [44]. “If only uniform pricing were to be allowed, any discussion of the combination ICI discounts would result in agreeing on the discounts for the monotherapy and all other uses of the backbone

treatment as well” [44]. Allowing IBP for combination therapies will, as a result, be critical to “avoid any breach of competition law when agreeing on the component prices for each combination therapy” [44].

Nevertheless, IBP could be challenging in practice since it requires information on volumes of usage by indication or assumptions of predicted utilization based on epidemiological data. It may be complex for many health systems to identify IBP use, as there should be no, or negligible opportunities for arbitrage [41]. Although there are a number of challenges in making IBP work in practice, it has the potential to be both advantageous and attainable for combination ICIs in treating various cancers, including NSCLC.

4. Discussion

Immunotherapy has revolutionized treatment for many cancers, including NSCLC. Combining ICIs either with each other or with other cancer drugs has improved responses, leading to the approval of various combination ICIs in NSCLC. However, the underlying mechanism for the improved clinical benefits of combination therapies compared with single-agent ICIs is not well studied [6]. The clinical success of ICI combinations has been widely interpreted as evidence of interaction-based mechanisms, but clinical trials do not always support such hypotheses. The superiority of combination ICI therapies to monotherapy in multiple indications does not indicate whether synergy or additivity is involved, or necessary [6]. Moreover, when component therapies are developed specifically to be used in combination with existing therapies, such therapies may not have been clinically evaluated independently outside of early-phase safety and dose escalation trials [8]. This poses additional challenges as the monotherapy effect of component therapies remains uncertain.

Palmer et al. reported that combination ICIs including atezolizumab may improve outcomes due to independent, rather than synergistic or additive, drug action [6]. In the IMpower150 clinical trial [NCT02366143], patients with metastatic NSCLC who received first-line treatment with atezolizumab plus chemotherapy and bevacizumab had longer PFS than would be expected by independent drug action, which suggests that this drug combination may have a synergistic or additive effect on clinical outcomes. However, Palmer et al. found that the “PFS of patients receiving the remaining evaluated combinations was similar to or shorter than the predicted outcomes for independent drug action, suggesting that the benefits of this combination were due to independent drug action rather than synergy or additivity” [6]. The key conclusion from this study was that “independent drug action could be sufficient to confer the PFS benefits required for a practice-changing therapy” [6]. As reported previously, improved benefits that can be predicted from monotherapy data has important implications for use in the design of future trials [45–47]. However, it remains to be seen to what extent the findings of Palmer et al. in NSCLC and various advanced solid tumors will “generalize to other neoplasms including hematologic malignancies, where drug additivity appears to be critical, [48] or to adjuvant therapies for early-stage cancers”. Nonetheless, these findings indicate that ICI combinations were clinically effective; but their effectiveness may be through a different mechanism of action. However, the key limitation of Palmer et al., is that it was a retrospective analysis involving imputed data [6]. An additional limitation was that “only a subset of existing combination therapies could be analyzed by their predictive model due to the lack of data on the monotherapy activity” [6].

From a valuation standpoint, progress may depend upon finding ways of aligning monotherapy drug prices with the value attributable to combination ICIs, which may be different from the value in another combination [49]. In recent years, potential value attribution methods have been suggested, [8,14]. We believe additional work will be needed to make improvements in these proposed methodologies, validate them and theoretically work on new ones.

In the identified conceptual frameworks, it was suggested that the

limitations in monotherapy data “tilt the scales” in favor of synergy [8]. This phenomenon may apply to the first-line nivolumab-ipilimumab combination therapy, where no proven single-agent ipilimumab activity exists. For example, some of the trials that the authors studied did not include an ICI monotherapy arm, requiring the use of data from another trial in a comparable patient population. In these cases, study authors matched the line of therapy and dosing to the greatest extent possible. Specifically, for the KEYNOTE-407 clinical trial [NCT02775435] in squamous NSCLC, monotherapy data were not available for pembrolizumab, but available for nivolumab, from the CheckMate-017 clinical trial [NCT01642004] [50]. A meta-analysis of 1’887 patients with NSCLC observed no significant difference in PFS or overall survival between pembrolizumab and nivolumab [51]. Therefore, the authors used nivolumab as a non-inferior comparator for pembrolizumab. This approach has the potential to “tilt the scales” in favor of synergy.

A distinct but important issue that relates to value attribution, is the issue of feasibility of negotiating a combined price for ICIs. It is important to recognize how manufacturers, who may be competitors in different indications, can share information in a compliant manner, without breaching competition law [44]. Although competition law differs by each jurisdiction, it usually does not permit competing manufacturers from discussing pricing with each other. While an agreed approach to value attribution could help, negotiation of combination drug prices remains to be the biggest challenge [8]. The manufacturer of an add-on therapy must devise a pricing strategy without knowing the pricing strategy of the manufacturer of the backbone therapy [52]. Flexibility in the pricing of the backbone therapy will depend on its current stage in the product life cycle as well as its current patent status and duration. If a backbone therapy has many years left under patent or holds a sufficiently large market share, its manufacturer may have little incentive to reduce its price [53,54].

When combination ICIs are made by different manufacturers, some form of IBP would be considered. This in turn requires reaching an agreement on the value, as well as the price of each specific use. Previous studies exploring the methods of value attribution indicated that the problem arises when the ‘backbone’ drug is priced at the limit (i.e., WTP threshold) of what a payer is prepared to pay given the outcomes delivered [8]. One feasible way forward to address the challenges of combination ICIs is adjusting the price of both constituents of a combination to reflect the value they each offer when used in that specific combination.

Lastly, it is essential that various stakeholders, including HTA bodies, manufacturers, policymakers, and payers are involved in the development of value attribution frameworks. Novel methods should provide principles that allow value attribution for most possible configurations of combination therapy including combinations of two or more drugs, and healthcare systems that use different measures of health gain to assess value and therefore price (i.e., for QALY or non-QALY based systems). To address these issues systematically, a structured approach should be used to tackle the key challenges of combination ICIs. Increased stakeholder involvement, at the regulatory, academic, industry, government, and research levels may help leverage resources for the advancement of valuation frameworks.

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Authors declare no conflicts of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the

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References

- [1] P. Darvin, S.M. Toor, V. Sasidharan Nair, et al., Immune checkpoint inhibitors: recent progress and potential biomarkers, *Exp. Mol. Med.* 50 (2018) 1–11, <https://doi.org/10.1038/s12276-018-0191-1>.
- [2] R.K. Vaddepally, P. Kharel, R. Pandey, et al., Review of Indications of FDA-Approved Immune Checkpoint Inhibitors per NCCN guidelines with the Level of Evidence, *Cancers* 12 (2020), E738, <https://doi.org/10.3390/cancers12030738>.
- [3] C. Schmidt, The benefits of immunotherapy combinations, *Nature* 552 (2017) S67–S69, <https://doi.org/10.1038/d41586-017-08702-7>.
- [4] K.M. Heinhuys, W. Ros, M. Kok, et al., Enhancing antitumor response by combining immune checkpoint inhibitors with chemotherapy in solid tumors, *Ann. Oncol.* 30 (2019) 219–235, <https://doi.org/10.1093/annonc/ndy551>.
- [5] S.C. Wei, C.R. Duffy, J.P. Allison, Fundamental mechanisms of immune checkpoint blockade therapy, *Cancer Discov.* 8 (2018) 1069–1086, <https://doi.org/10.1158/2159-8290.CD-18-0367>.
- [6] A.C. Palmer, B. Izar, H. Hwangbo, et al., Predictable clinical benefits without evidence of synergy in trials of combination therapies with immune-checkpoint inhibitors, *Clin. Cancer Res.* 28 (2022) 368–377, <https://doi.org/10.1158/1078-0432.CCR-21-2275>.
- [7] R. Zaim, K. Redekop, C. Uyl-De Groot, How should value be attributed to the constituent parts of combination immunotherapy in non-small cell lung cancer? *Value Health* 23 (2020) S458–S459, <https://doi.org/10.1016/j.jval.2020.08.339>.
- [8] A. Briggs, A. Doyle, J. Schneider, An Attribution of Value Framework for Combination Therapies. Report by the Value Attribution Working Group, 2021. (<https://www.takeda.com/49a844/siteassets/en-gb/home/what-we-do/combination-treatments/a-value-attribution-framework-for-combination-therapies-takeda-whitepaper.pdf>).
- [9] Addressing Challenges in Access to Oncology Medicines. Analytical Report. OECD. (<https://www.oecd.org/health/health-systems/Addressing-Challenges-in-Access-to-Oncology-Medicines-Analytical-Report.pdf>) (Accessed 15 June 2022).
- [10] S. Davis, Assessing technologies that are not cost-effective at a zero price. NICE Decision Support Unit Report, 2014. (https://www.ncbi.nlm.nih.gov/books/NBK310371/pdf/Bookshelf_NBK310371.pdf) (Accessed 1 Jun 2022).
- [11] S. Davis, R. Akehurst, How do we evaluate technologies that are not cost effective at a zero price? *Value Outcomes Spotlight* (2016) 6–8.
- [12] N.R. Latimer, A. Towse, C. Henshall, Not cost-effective at zero price: valuing and paying for combination therapies in cancer, *Expert Rev. Pharm. Outcomes Res* 21 (2021) 331–333, <https://doi.org/10.1080/14737167.2021.1879644>.
- [13] N.R. Latimer, D. Pollard, A. Towse, et al., Challenges in valuing and paying for combination regimens in oncology: reporting the perspectives of a multi-stakeholder, international workshop, *BMC Health Serv. Res* 21 (2021) 412, <https://doi.org/10.1186/s12913-021-06425-0>.
- [14] A. Towse, M. Lothgren, L. Steuten, et al., Why we need a new outcomes-based value attribution framework for combination regimens in oncology. OHE Consulting Report, London. Office of Health Economics, 2021. (<https://www.ohe.org/publications/why-we-need-new-outcomes-based-value-attribution-framework-combination-regimens>) (Accessed 2 January 2022).
- [15] The US Food and Drug Administration: Pembrolizumab, 2017. (<https://www.fda.gov/drugs/resources-information-approved-drugs/pembrolizumab-keytruda-5-10-2017>) (Accessed 1 June 2022).
- [16] The US Food and Drug Administration grants regular approval for pembrolizumab in combination with chemotherapy for first-line treatment of metastatic nonsquamous NSCLC, 2018. (<https://www.fda.gov/drugs/resources-information-approved-drugs/fda-grants-regular-approval-pembrolizumab-combination-chemotherapy-first-line-treatment-metastatic>) (Accessed 1 June 2022).
- [17] The US Food and Drug Administration approves pembrolizumab in combination with chemotherapy for first-line treatment of metastatic squamous NSCLC, 2018. (<https://www.fda.gov/drugs/fda-approves-pembrolizumab-combination-chemotherapy-first-line-treatment-metastatic-squamous-nsclc>) (Accessed 1 June 2022).
- [18] The US Food and Drug Administration approves atezolizumab with chemotherapy and bevacizumab for first-line treatment of metastatic non-squamous NSCLC, 2018. (<https://www.fda.gov/drugs/fda-approves-atezolizumab-chemotherapy-and-bevacizumab-first-line-treatment-metastatic-non-squamous>) (Accessed 1 June 2022).
- [19] The US Food and Drug Administration approves atezolizumab with nab-paclitaxel and carboplatin for metastatic NSCLC without EGFR/ALK aberrations, 2019. (<https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-atezolizumab-nab-paclitaxel-and-carboplatin-metastatic-nsclc-without-egfralk>) (Accessed 1 June 2022).
- [20] The US Food and Drug Administration approves nivolumab plus ipilimumab for first-line mNSCLC (PD-L1 tumor expression $\geq 1\%$), 2020. (<https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-nivolumab-plus-ipilimumab-first-line-mnscld-pd-l1-tumor-expression-1>) (Accessed 1 June 2022).
- [21] The US Food and Drug Administration. FDA Approves Nivolumab Plus Ipilimumab and Chemotherapy for First-Line Treatment of Metastatic NSCLC, 2020. (<https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-nivolumab-plus-ipilimumab-and-chemotherapy-first-line-treatment-metastatic-nsclc>) (Accessed 14 January 2022).
- [22] The US Food and Drug Administration approves neoadjuvant nivolumab and platinum-doublet chemotherapy for early-stage non-small cell lung cancer. 2022. (<https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-neoadjuvant-nivolumab-and-platinum-doublet-chemotherapy-early-stage-non-small-cell-lung-cancer>) (Accessed 1 June 2022).
- [23] M.A. Socinski, R.M. Jotte, F. Cappuzzo, et al., Atezolizumab for first-line treatment of metastatic nonsquamous NSCLC, *New Engl. J. Med.* 378 (2018) 2288–2301, <https://doi.org/10.1056/NEJMoa1716948>.
- [24] A. Rittmeyer, F. Barlesi, D. Waterkamp, et al., Atezolizumab versus docetaxel in patients with previously treated non-small-cell lung cancer (OAK): a phase 3, open-label, multicentre randomised controlled trial, *Lancet* 389 (2017) 255–265, [https://doi.org/10.1016/S0140-6736\(16\)32517-X](https://doi.org/10.1016/S0140-6736(16)32517-X).
- [25] S. Peters, S. Gettinger, M.L. Johnson, et al., Phase II trial of atezolizumab as first-line or subsequent therapy for patients with programmed death-ligand 1-selected advanced non-small-cell lung cancer (BIRCH), *J. Clin. Oncol.* 35 (2017) 2781–2789, <https://doi.org/10.1200/JCO.2016.71.9476>.
- [26] L. Dron, S. Golchi, G. Hsu, et al., Minimizing control group allocation in randomized trials using dynamic borrowing of external control data - an application to second line therapy for non-small cell lung cancer, *Conte Clin. Trials Commun.* 16 (2019), 100446, <https://doi.org/10.1016/j.conctc.2019.100446>.
- [27] L.E. Schnipper, N.E. Davidson, D.S. Wollins, et al., Updating the American society of clinical oncology value framework: revisions and reflections in response to comments received, *J. Clin. Oncol.* 34 (2016) 2925–2934, <https://doi.org/10.1200/JCO.2016.68.2518>.
- [28] NCCN Clinical Practice Guidelines in Oncology with NCCN Evidence Blocks.
- [29] N.I. Cherny, R. Sullivan, U. Dafni, et al., A standardised, generic, validated approach to stratify the magnitude of clinical benefit that can be anticipated from anti-cancer therapies: the European Society for Medical Oncology Magnitude of Clinical Benefit Scale (ESMO-MCBS), *Ann. Oncol.* 26 (2015) 1547–1573, <https://doi.org/10.1093/annonc/ndv249>.
- [30] Drug Pricing Lab. DrugAbacus Methods. (<https://www.drugpricinglab.org/tools/drug-abacus/methods/>) (Accessed 1 June 2022).
- [31] L. Paz-Ares, T.-E. Ciuleanu, M. Cobo, et al., First-line nivolumab plus ipilimumab combined with two cycles of chemotherapy in patients with non-small-cell lung cancer (CheckMate 9LA): an international, randomised, open-label, phase 3 trial, *Lancet Oncol.* 22 (2021) 198–211, [https://doi.org/10.1016/S1470-2045\(20\)30641-0](https://doi.org/10.1016/S1470-2045(20)30641-0).
- [32] M.D. Hellmann, L. Paz-Ares, R. Bernabe Caro, et al., Nivolumab plus Ipilimumab in advanced non-small-cell lung cancer, *New Engl. J. Med.* 381 (2019) 2020–2031, <https://doi.org/10.1056/NEJMoa1910231>.
- [33] ClinicalTrials.gov: Phase II Study for Previously Untreated Subjects With Non Small Cell Lung Cancer or Small Cell Lung Cancer, 2018. (<https://clinicaltrials.gov/ct2/show/NCT00527735>).
- [34] X. Wan, X. Zeng, L. Peng, et al., Cost-effectiveness analysis of nivolumab plus ipilimumab for advanced non-small-cell lung cancer, *Front. Pharmacol.* 12 (2021), 580459, <https://doi.org/10.3389/fphar.2021.580459>.
- [35] P.T. Courtney, A.T. Yip, D.R. Cherry, et al., Cost-effectiveness of Nivolumab-Ipilimumab combination therapy for the treatment of advanced non-small cell lung cancer, *JAMA Netw. Open* 4 (2021), e218787, <https://doi.org/10.1001/jamanetworkopen.2021.8787>.
- [36] H. Hu, L. She, M. Liao, et al., Cost-effectiveness analysis of nivolumab plus ipilimumab vs. chemotherapy as first-line therapy in advanced non-small cell lung cancer, *Front. Oncol.* 10 (2020) 1649, <https://doi.org/10.3389/fonc.2020.01649>.
- [37] Y. Peng, X. Zeng, L. Peng, et al., Cost-effectiveness of nivolumab plus ipilimumab combined with two cycles of chemotherapy as first-line treatment in advanced non-small cell lung cancer, *Adv. Ther.* 38 (2021) 3962–3972, <https://doi.org/10.1007/s12325-021-01788-6>.
- [38] X. Hao, A. Shen, B. Wu, Cost-effectiveness of nivolumab plus ipilimumab as first-line therapy in advanced non-small-cell lung cancer, *Front. Pharmacol.* 12 (2021), 573852, <https://doi.org/10.3389/fphar.2021.573852>.
- [39] J. Li, T. Zhang, Y. Xu, et al., Cost-effectiveness analysis of nivolumab plus ipilimumab versus chemotherapy as first-line treatment in advanced NSCLC, *Immunotherapy* 12 (2020) 1067–1075, <https://doi.org/10.2217/imt-2020-0112>.
- [40] S.-C. Yang, N. Kunst, C.P. Gross, et al., Cost-effectiveness of nivolumab plus ipilimumab with and without chemotherapy for advanced non-small cell lung cancer, *Front. Oncol.* 11 (2021), 760686, <https://doi.org/10.3389/fonc.2021.760686>.
- [41] D. Dankó, J.-Y. Blay, L.P. Garrison, Challenges in the value assessment, pricing and funding of targeted combination therapies in oncology, *Health Policy* 123 (2019) 1230–1236, <https://doi.org/10.1016/j.healthpol.2019.07.009>.
- [42] A. Chandra, C. Garthwaite, The economics of indication-based drug pricing, *New Engl. J. Med.* 377 (2017) 103–106, <https://doi.org/10.1056/NEJMp1705035>.
- [43] U. Persson, J.M. Norlin, Multi-indication and combination pricing and reimbursement of pharmaceuticals: opportunities for improved health care through faster uptake of new innovations, *Appl. Health Econ. Health Policy* 16 (2018) 157–165, <https://doi.org/10.1007/s40258-018-0377-7>.
- [44] T. Podkonjak, H. Taylor, B. Morgan et al. Voluntary Arbitration Framework for Combination Therapies. A proposed process by the Voluntary Arbitration Working Group, 2021. (<https://www.takeda.com/4a81d5/siteassets/en-gb/home/what-we-do/combination-treatments/voluntary-arbitration-framework-for-combination-therapies-takeda-whitepaper-september2021.pdf>) (Accessed 1 Jun 2022).
- [45] C. Chen, F. Liu, Y. Ren, et al., Independent drug action and its statistical implications for development of combination therapies, *Conte Clin. Trials* 98 (2020), 106126, <https://doi.org/10.1016/j.cct.2020.106126>.
- [46] L.Z. Sun, C. Wu, X. Li, et al., Independent action models and prediction of combination treatment effects for response rate, duration of response and tumor size change in oncology drug development, *Conte Clin. Trials* 106 (2021), 106434, <https://doi.org/10.1016/j.cct.2021.106434>.

- [47] E.V. Schmidt, M.J. Chisamore, M.F. Chaney, et al., Assessment of clinical activity of PD-1 checkpoint inhibitor combination therapies reported in clinical trials, *JAMA Netw. Open* 3 (2020), e1920833, <https://doi.org/10.1001/jamanetworkopen.2019.20833>.
- [48] A.C. Palmer, C. Chidley, P.K. Sorger, A curative combination cancer therapy achieves high fractional cell killing through low cross-resistance and drug additivity, *Elife* 8 (2019), e50036, <https://doi.org/10.7554/eLife.50036>.
- [49] C. Henshall, A. Towse, Time to Tackle the Challenges around Combination Therapies for Cancer, 2021. (<https://www.ohe.org/news/time-tackle-challenges-around-combination-therapies-cancer>).
- [50] J. Brahmer, K.L. Reckamp, P. Baas, et al., Nivolumab versus docetaxel in advanced squamous-cell non-small-cell lung cancer, *New Engl. J. Med.* 373 (2015) 123–135, <https://doi.org/10.1056/NEJMoa1504627>.
- [51] T.-R. Peng, F.-P. Tsai, T.-W. Wu, Indirect comparison between pembrolizumab and nivolumab for the treatment of non-small cell lung cancer: a meta-analysis of randomized clinical trials, *Int. Immunopharmacol.* 49 (2017) 85–94, <https://doi.org/10.1016/j.intimp.2017.05.019>.
- [52] R. Bayat Mokhtari, T.S. Homayouni, N. Baluch, et al., Combination therapy in combating cancer, *Oncotarget* 8 (2017) 38022–38043, <https://doi.org/10.18632/oncotarget.16723>.
- [53] K.N. Vokinger, T.J. Hwang, P. Daniore, et al., Analysis of launch and postapproval cancer drug pricing, clinical benefit, and policy implications in the US and Europe, *JAMA Oncol.* 7 (2021), e212026, <https://doi.org/10.1001/jamaoncol.2021.2026>.
- [54] A.W. Mulcahy, C.M. Whaley, M. Gizaw, et al., International Prescription Drug Price Comparisons: Current Empirical Estimates and Comparisons with Previous Studies, RAND Corporation, Santa Monica, CA, 2021, <https://doi.org/10.7249/RR2956>.